RAKESH KUMAR

Mechanical Engineer | CFD & Thermal-Fluid Systems Specialist

Post-Doctoral Fellow, Aerospace Engineering, IIT Bombay

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PROFESSIONAL SUMMARY

CFD and thermal-fluid engineer with 7+ years of experience in aerodynamics, thermal-fluid, and structural analysis for turbomachinery, hydrokinetic and wind turbines, rotorcraft systems, and HVAC applications. Strong expertise in ANSYS Fluent, ANSYS CFX, OpenFOAM, and ANSYS Structural, with hands-on experience in rotor aerodynamics, rotating machinery CFD, FSI, heat transfer, HVAC airflow modeling, dynamic and overset meshing, UDFs, erosion modeling, and FEA. Proven ability to execute high-fidelity transient CFD and coupled CFD–FEA simulations on HPC systems for industry R&D, translating results into engineering-ready design insights for performance, reliability, and energy efficiency. PhD from IIT(ISM) Dhanbad.

PROFESSIONAL & RESEARCH EXPERIENCE

Post-Doctoral Fellow

Indian Institute of Technology Bombay, Mumbai, India

Nov 2024 - Present

- Performed high-fidelity aerodynamic CFD analysis of advanced rotorcraft configurations, including cyclorotors and tandem rotors, using ANSYS Fluent, focusing on thrust generation, wake interaction, and unsteady flow behavior.
- Developed and implemented User-Defined Functions (UDFs) for time-varying blade kinematics and complex flow physics to improve simulation accuracy.
- Applied dynamic, overset, and sliding mesh techniques to simulate rotating machinery motion and rotor-wake interactions.
- Executed HPC-based parallel CFD simulations and post-processed aerodynamic loads, pressure distribution, and flow structures for performance optimization.

Senior Researcher

Indian Institute of Technology (Indian School of Mines) Dhanbad, Dhanbad, India

May 2024 - Nov 2024

- Wind Turbine R&D, led the design and numerical simulation of a utility-scale wind turbine model using ANSYS CFX for performance analysis and ANSYS Static Structural for stress validation. Project focused on developing a predictive condition monitoring system for wind turbines using simulation data.

Senior Research Fellow

Indian Institute of Technology (Indian School of Mines) Dhanbad, Dhanbad, India

Oct 2019 - Nov 2022

- Hydraulic Turbine R&D, conducted a high-fidelity erosion and wear prediction model for radial hydraulic turbines using CFD–DEM coupling (ANSYS Fluent + EDEM). Analyzed the influence of particle size, shape, concentration, and flow rate on turbine component erosion.

TECHNICAL SKILLS

- CFD software: ANSYS CFX & FLUENT, OpenFoam, Dynamic and Overset mesh, Fluent Meshing (Watertight Geometry Workflow), User-Defined Function (UDF): (Turbomachines, Aerodynamics, Internal flows, Multiphase flow, Thermal-fluid analysis, Wear and Erosion modelling)
- Thermal & Heat Transfer Modeling: Heat transfer, thermal management, and HVAC airflow analysis.
- CAD Software: SolidWorks, SpaceClaim, Siemens NX, BladeGen, AutoCAD
- Finite Element Analysis (FEA) and Fluid-Solid Interaction (FSI): ANSYS Workbench (Static and Transient Structural: Linear & Non-Linear Analysis), CFD–FEA coupling
- CAE Pre-processing: ANSA (geometry cleanup, surface and volume mesh generation).
- Computational Fluid Dynamics (CFD)-Discrete Element Method (DEM) modelling: ANSYS Fluent–EDEM coupling
- EDEM software: Bulk and Granular material simulation (Wear & Erosion)
- HPC & Parallel Computing: Linux-based HPC clusters, MPI parallelization, multi-node Fluent/CFX simulations, batch job submission, solver performance optimization
- Coding: Python, MATLAB, C, UDF scripting, CEL (CFX Expression Language)
- Post-Processing: Design optimization and parametric CFD studies, CFD-Post, Tecplot, ParaView, Excel, OriginPro.

SOFT SKILLS

• Leadership • Communication • Problem Solving • Adaptability

AREAS OF EXPERTISE

Turbomachinery, thermal-fluid performance of rotating systems, aerodynamics, CFD, FSI, fluid dynamics, flow and thermal analysis, multiphase flow, FEA, CFD-DEM coupling, erosion modeling, thermal management systems, heat transfer, and HVAC systems.

EDUCATION QUALIFICATIONS

- **Doctor of Philosophy (Ph.D.)**, Mechanical Engineering
  Indian Institute of Technology (Indian School of Mines) Dhanbad, Dhanbad, India
  July 2018 - April 2024 || ***Date of Awarded: 04 April 2024***
  Thesis title: *Performance analysis of Darrieus hydrokinetic turbine in different flow conditions*.
- **Master of Technology (M.Tech.)**, Mechanical Engineering (Thermal Engineering)
  Indian Institute of Technology (Indian School of Mines) Dhanbad, Dhanbad, India.
  August 2016 - May 2018
  Thesis title: *Effect of number of blades on performance of helical Darrieus hydrokinetic turbine.*
- **Bachelor of Technology (B.Tech.)**, Mechanical Engineering
  Maharana Pratap Engineering College Kanpur, AKTU Lucknow, India.
  July 2010 - June 2014
  Thesis title: *Design analysis and processes involved in manufacturing of a Bicycle*.

ACHIEVEMENTS

- **Award:** Inder Mohan Thapar Research (IMTR) award for the year (2023 & 2024)—Awarded by Indian Institute of Technology (Indian School of Mines), Dhanbad, India for excellence in research publications during the calendar years 2022 and 2023.
- **Patent:** Kumar, R. and Sarkar, S., "An In-Pipe Hydrokinetic Turbine (IPHKT) Structure" (Indian patent number 202331002558), Publication Date: 20/01/2023.

PUBLICATIONS IN PEER-REVIEWED SCI JOURNALS/BOOK CHAPTERS/CONFERENCES

- Kumar, R., & Sarkar, S. (2025). Performance Evaluation of Modified Hydrokinetic Turbines in Pipe Flow with Velocity Correction Technique. Renewable Energy, 242, 122485. https://doi.org/10.1016/j.renene.2025.122485 (Impact Factor: 9.1).
- Kumar, R., Nag, A. K., & Sarkar, S. (2024). Performance analysis of spherically curbed hydrokinetic turbine arranged in ln-line array in a closed conduit. Renewable Energy, 232, 121110. https://doi.org/10.1016/j.renene.2024.121110 (Impact Factor: 9.1).
- Kumar, R., Yadav, I., & Sarkar, S. (2024). Performance and Stress Analysis of Helical Darrieus Hydrokinetic Turbine. In International Conference on Industrial Problems on Machines and Mechanism (pp. 567-576). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-99-4270-1_56.
- Kumar, R., & Sarkar, S. (2023). Performance analysis of spherical-shaped Darrieus hydrokinetic turbine for an in-pipe hydropower system. Energy Conversion and Management, 294, 117600. https://doi.org/10.1016/j.enconman.2023.117600 (Impact Factor: 10.9).
- Kumar, R., and Sarkar, S. (2023). Erosion analysis of radial flow hydraulic turbine components through FLUENT-EDEM coupling. Powder Technology, 428, 118800. https://doi.org/10.1016/j.powtec.2023.118800 (Impact Factor: 4.6).
- Sarkar, S., Nag, A.K. and Kumar, R., (2023). "An overview of small-scale hydropower and its recent development". Renewable Energy Production and Distribution Volume 2, pp.249-314. https://doi.org/10.1016/B978-0-443-18439-0.00009-4.
- Kumar, R., & Sarkar, S. (2022). Effect of design parameters on the performance of helical Darrieus hydrokinetic turbines. Renewable and Sustainable Energy Reviews, 162, 112431. https://doi.org/10.1016/j.rser.2022.112431 (Impact Factor: 16.3).

POSITIONS AND RESPONSIBILITY

- **Teaching Assistant**— Low Speed Aerodynamics and Rotary Wing Aerodynamics at IIT Bombay (Jan 2025 to present).
- **Teaching and Lab Assistant**— Turbomachinery, Fluid Mechanics, Fluid Machines, Engineering Mechanics, Engineering Graphics, and Theory of Machines at IIT (ISM) Dhanbad (August 2018 - April 2023).
- **Vice-President**, NGO "Karmajyoti", IIT (ISM) Dhanbad (Jan 2022 – Dec 2022); Volunteer (Jul 2018 – Dec 2022).
- **Member**, Hostel Executive Committee, Emerald Hostel, IIT (ISM) Dhanbad (2021–22).